

AMIRIS Input Data Requirements

Request for cross-reference against the department's Energy Brainpool data holdings

Prepared by: Muideen Oladayo Ajiboye | 20 July 2026

Purpose

To test whether AMIRIS (DLR's open-source electricity market model) can reproduce the same hourly price results as the department's purchased Energy Brainpool data, we plan to run AMIRIS using the same underlying assumptions Brainpool used for a given year (e.g. 2019), rather than independently sourced data. This isolates whether any difference in results comes from AMIRIS's internal modelling logic, rather than from using different input data - a cleaner comparison than testing against independently sourced data.

Please review the data items below against what the Brainpool package contains, and indicate for each one whether an equivalent is available, partially available, or not available. Where something is not available from Brainpool, AMIRIS's own independently-sourced data will be used instead, and this will be documented as a methodological limitation of the comparison.

How AMIRIS Uses This Data (in brief)

AMIRIS simulates the German day-ahead electricity market hour by hour for a full year. It creates a virtual agent for each type of market participant (each conventional fuel type, each renewable technology, storage operators, and demand), each bidding into a simulated auction using the data below. The market clears each hour by accepting the cheapest offers first until enough electricity is queued up to meet demand; the price of the last accepted offer becomes that hour's simulated electricity price - the same mechanism the real day-ahead auction uses. The output is therefore mainly a full year of hourly electricity prices (8,760 values).

A. Electricity Demand

In plain terms:

Simply how much electricity all of Germany used, hour by hour, across the year - like a giant, constantly-updating electricity meter reading for the entire country instead of one house.

Data item	What it means, and why AMIRIS needs it	Needed resolution
Hourly load	Total German electricity demand to be served each hour. AMIRIS needs this because the simulated market has to work out how much electricity to produce to cover it, every single hour.	Hourly (8,760 values/year)

B. Conventional (Fuel-Burning) Power Plant Fleet

In plain terms:

This section is about the coal, gas, oil and nuclear power plants - the ones that burn or split fuel to make electricity. Needed separately for each of: Nuclear, Lignite (brown coal), Hard coal, Natural gas, Oil.

Data item	What it means, and why AMIRIS needs it	Needed resolution
Installed capacity	The total 'horsepower' of that fuel type's fleet, in megawatts (MW) - e.g. all coal plants added together. Sets the ceiling on how much that fuel type could ever produce.	One value per year (or per change in capacity)
Efficiency range	How much of the energy locked inside the fuel actually becomes electricity, versus being lost as waste heat - a 40%-efficient plant turns 40 out of every 100 units of fuel energy into electricity.	One value (or range) per fuel type
Typical block size	How big a single, individual plant of this type usually is. AMIRIS models plants as separate units, not one giant blob of capacity, so it needs to know how to split the total into realistic individual plants.	One value per fuel type
Variable operating cost	The extra cost of running the plant for one more hour, beyond the fuel itself - things like wear and tear. Expressed as EUR per MWh produced.	One value per fuel type
CO2 emissions factor	How much carbon dioxide is released for every unit of electricity that fuel type produces - coal releases much more than gas, for example.	One value per fuel type
Outage/availability	What fraction of that fleet is broken down or shut off for repairs at any given time - power plants aren't available 100% of the time, similar to how a car needs regular servicing.	Hourly if available; monthly/seasonal acceptable

C. Renewable Power Plant Fleet

In plain terms:

This section is about solar, wind, biomass and hydro power. Needed separately for each of: Solar (PV), Wind onshore, Wind offshore, Biomass, Run-of-river hydro, Other renewables.

Data item	What it means, and why AMIRIS needs it	Needed resolution
Installed capacity	The total maximum possible output if the sun were shining or the wind were blowing at its absolute best, in megawatts. Ideally split by subsidy vintage/scheme if available, since older and newer installations are often paid differently.	One value per year (or per change in capacity)
Generation profile	How much of that maximum was ACTUALLY produced, hour by hour - since a solar panel makes nothing at night and a wind turbine makes little on a calm day, this is the real, weather-driven output shape, not just the capacity number. This is the item most likely to need clarifying - see note below.	Hourly (8,760 values/year) - the single most important item to confirm
Subsidy/support scheme	Whether and how the government pays renewable-plant owners extra money on top of what they earn selling electricity (a fixed guaranteed price, or a top-up on the market price) - this affects how cheaply owners are willing to sell.	One value per technology/vintage

Note on generation profiles: this is the item most likely to be a gap. If Brainpool's data consists of capacity totals only (not an hourly generation shape), AMIRIS's own independently-sourced weather-driven profiles will be used to convert those capacity totals into hourly output. Please indicate specifically whether an hourly (or at least monthly/seasonal) shape is included, separately from the capacity totals themselves.

D. Storage (Batteries, Pumped Hydro, etc.)

In plain terms:

This section is about anything that can store electricity and release it later, such as batteries or pumped-hydro reservoirs.

Data item	What it means, and why AMIRIS needs it	Needed resolution
Power and energy capacity	Power (MW) is how fast it can charge or discharge - like a tap's flow rate. Energy (MWh) is how much it can hold in total - like the size of the bucket underneath the tap.	One value per year
Round-trip efficiency	How much of the electricity put INTO storage comes back OUT again - some is always lost, the same way a rechargeable battery never gives back exactly what you put in.	One value
Number of operators	Whether storage is modelled as one big combined pool, or as several separate operators competing against each other, if known.	One value, if available

E. Fuel and Carbon Prices

In plain terms:

What it costs power plants to buy the raw fuel they burn, and what it costs them to be allowed to emit each tonne of CO₂ - both directly affect how expensive it is to run a plant, and therefore what price it needs to sell electricity at to break even. These move slowly, so monthly or even quarterly figures are sufficient - hourly resolution is not needed.

Data item	What it means	Needed resolution
Hard coal price	Cost of coal delivered to the power plant, per MWh of energy content in the coal (not per tonne).	Monthly or quarterly
Natural gas price	Cost of gas delivered to the power plant, per MWh of energy content.	Monthly or quarterly
Oil price	Cost of oil delivered to the power plant, per MWh of energy content.	Monthly or quarterly
CO ₂ allowance price	The market price of a permit allowing a plant to legally emit one tonne of CO ₂ , under the EU's carbon trading scheme (EU ETS).	Monthly or quarterly

F. Structural / Methodology Questions

These are not raw data items, but assumptions built into how Brainpool's model works that materially affect the comparison. Please clarify each if possible:

- Cross-border trade: did Brainpool's model allow electricity import/export with neighbouring countries, or treat Germany as a standalone market? (AMIRIS, as currently configured, treats Germany as standalone - this is a known, confirmed source of price difference in our earlier backtesting work.)
- Bidding behaviour: were power plants assumed to bid strictly at production cost, or with a strategic markup/margin?
- Negative prices: does Brainpool's output include negative electricity prices (a real, regularly observed feature of the German market), and if so, roughly how often?

Next Steps

Once the above is reviewed, we will build an AMIRIS scenario for the agreed year using Brainpool's assumptions wherever supplied, filling any remaining gaps with AMIRIS's own independently-sourced real data (documented explicitly). We will then run AMIRIS and compare its simulated hourly price against both Brainpool's price output and the real historical price for that year, giving a three-way comparison. Close agreement between AMIRIS and Brainpool would support using AMIRIS as a free substitute for future years; a divergence would be investigated to identify the specific modelling difference responsible.